MODULE 20

GPS, Cell IDs and IMEI

Certified Cyber Criminologist - Police / LEA Training

Samir Raja (CYBER07) | Cyber Crime Investigator & Offensive Security Researcher

What You Will Learn

Your phone **keeps telling where you are** - every call, every map search. That diary becomes your **strongest investigative weapon**.

- Three routes to a phone's location: **GPS, mobile tower, WiFi**
- **Cell ID, LAC, CID** - how a tower estimates location
- Reading a **CDR**: top contacts, night patterns, crime windows
- **IMEI, IMSI, MSISDN, ICCID** - four identities, four jobs
- Tower-dump filtering, the **CEIR portal**, and the **legal process** for it all

Your Phone Is a Location Diary

Every call connects to a tower. Every map search fixes a position. **Convenience for you - evidence for the investigator.**

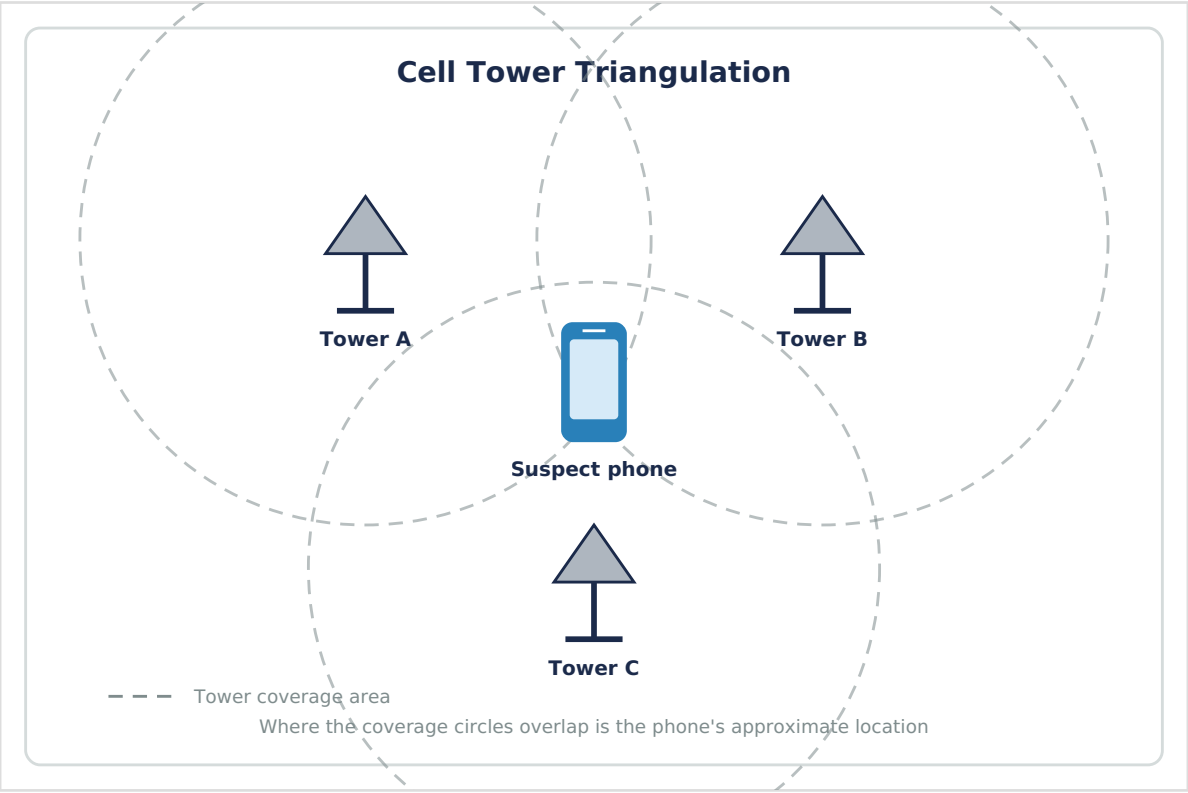
- **GPS:** satellites fix the phone within metres - but the phone keeps GPS off to save battery
- **Mobile tower:** the phone is always on some tower - the tower logs which phone is in its range
- **WiFi:** connecting to a WiFi records that WiFi's location too
- All three together tell the location story - **one covers the other's gaps**

Cell ID, LAC and CID

Every tower has an identity - **learn to read it like an address**:

- **Cell ID:** the tower's own identity - your number's tower connection is logged with time
- **LAC** (Location Area Code): the area of a group of towers
- **CID** (Cell Identity): one tower - or one of its sectors - inside that area
- Cities: dense towers, **fairly accurate**. Villages: sparse towers, rough estimate - **both useful**

How Towers Locate a Phone



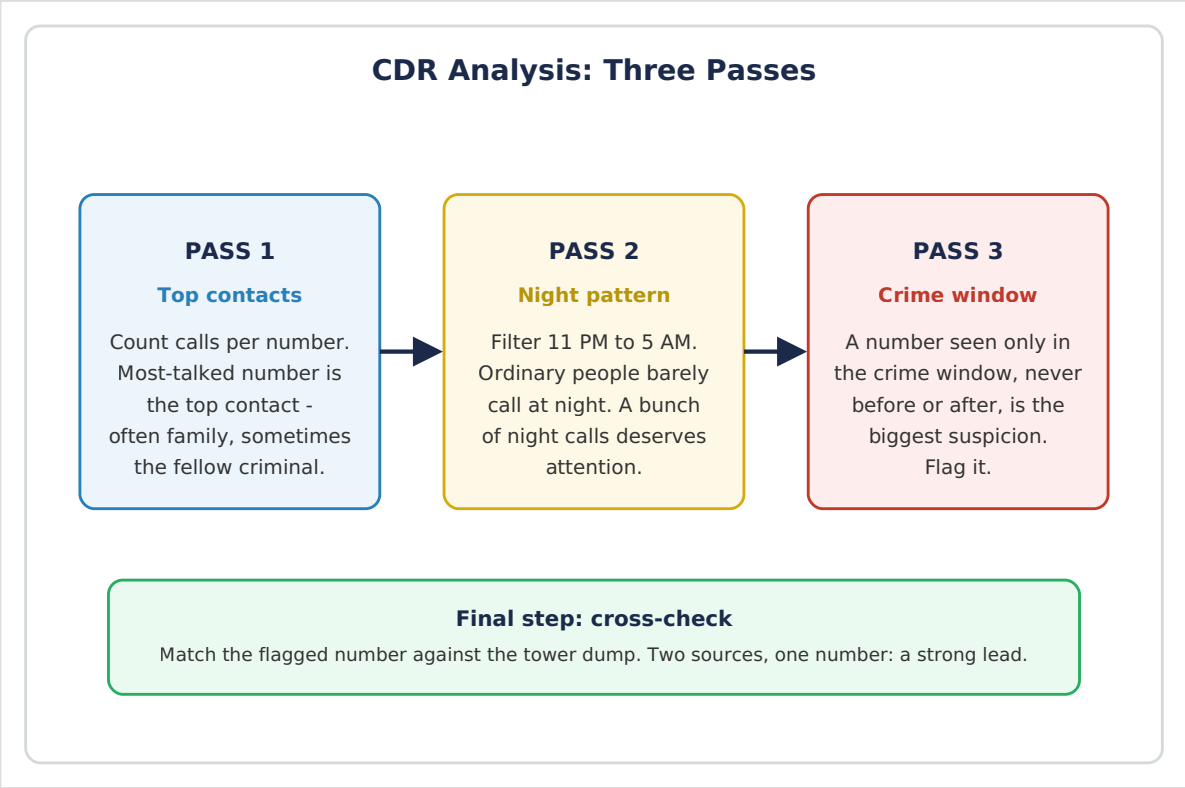
- Phone to tower: **which tower, at what time**
- At the crime hour, the accused's tower **is a strong link**
- Combine with CDR for the **full picture**

CDR: The Call Account Book

CDR - Call Detail Record - is the account of **every call and SMS**, kept by the telecom company:

- Who called whom, **how long**, at **what time** - and which tower the phone used
- The investigator reads it **like a map**: contacts, timing, towers
- Three passes unlock it: **top contacts, night pattern, crime window**
- A CDR is private data - **legal process only**, always

CDR Analysis: Three Passes



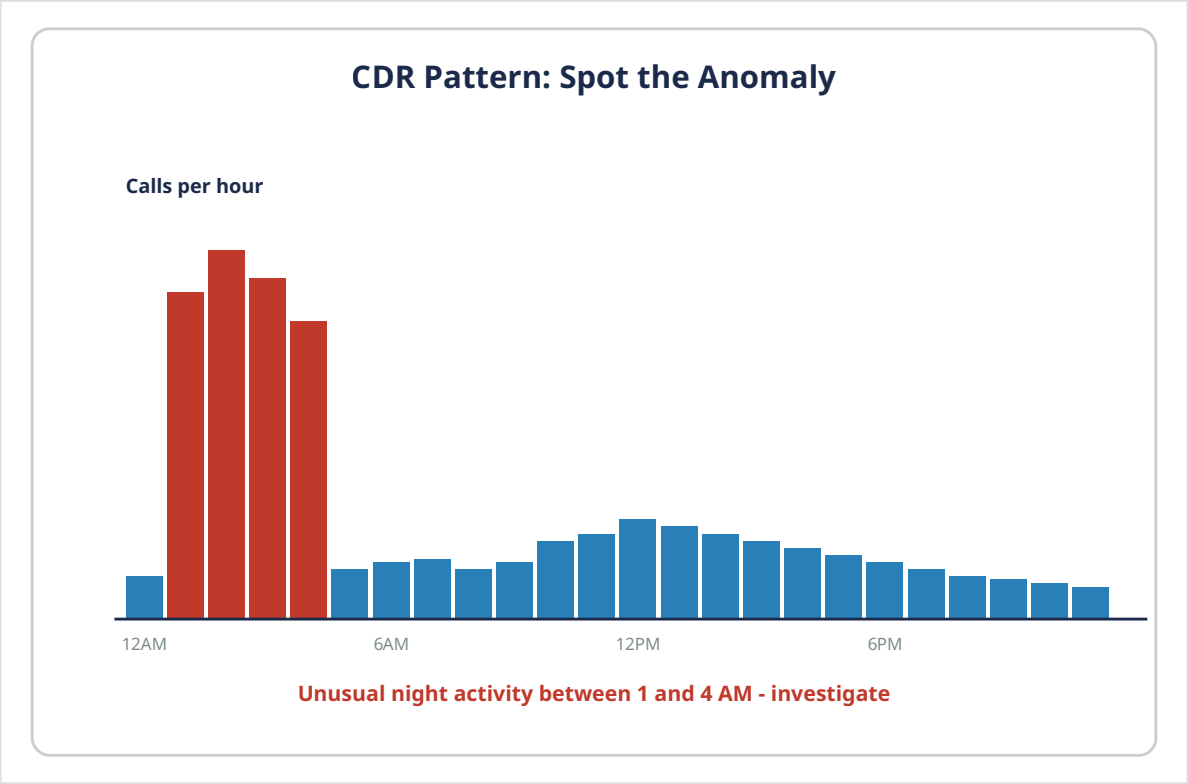
- Pass 1 counts, Pass 2 filters, Pass 3 **flags**
- The crime-window-only number is your **prime lead**
- Cross-check with the tower dump - **two sources, one number**

Reading a CDR in a Spreadsheet

Telecom companies give CDRs as **CSV or Excel**. LibreOffice Calc - free - does the whole job:

- **Sort** the date-time column A to Z - the story must run in time order
- **AutoFilter** 11 PM to 5 AM - the night calls surface instantly
- **COUNTIF** counts a number's appearances - **=COUNTIF(B2:B100,"98111 00111")** finds the top contact
- Golden rule: **never touch the original file** - analyse a copy, seal the original with its hash

The Night Pattern



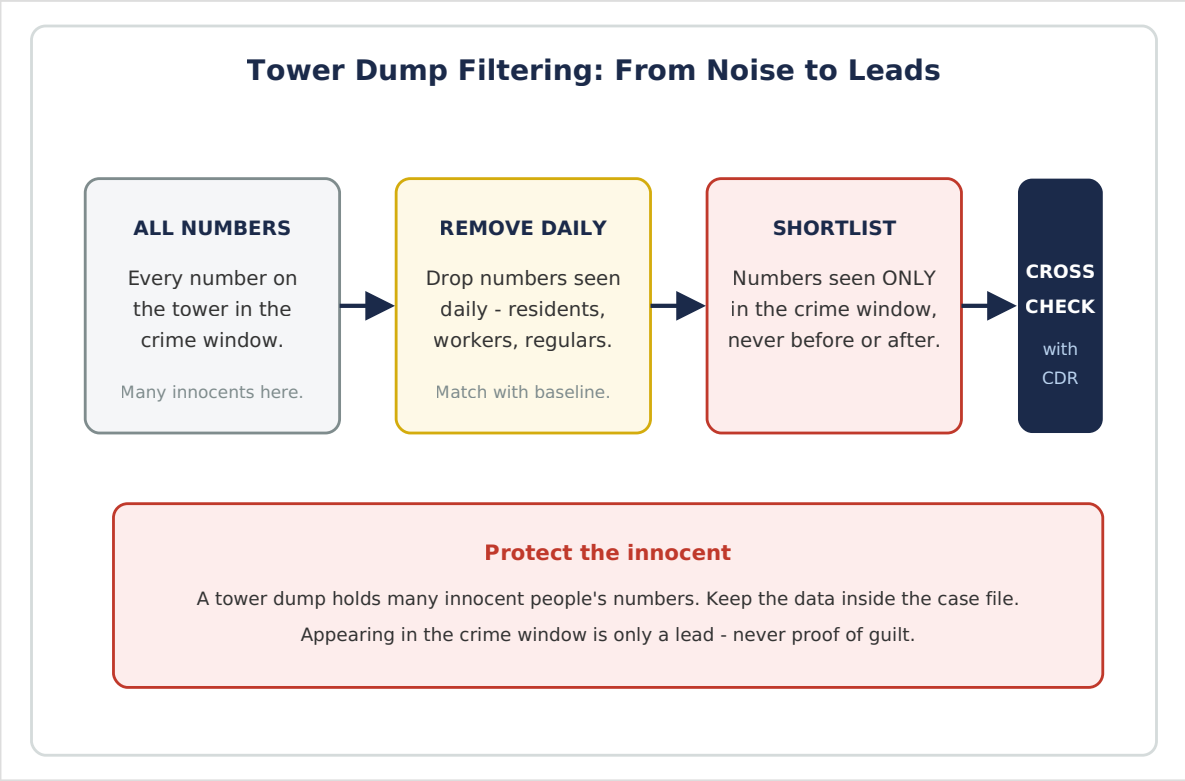
- Red bars at 1 to 4 AM: **not normal behaviour**
- Night calls plus a **different tower** - the suspect moved
- Day towers vs night towers **draw the map**

Tower Dumps: A Whole Place, One List

Sometimes you need **every phone in an area**, not one number. That list is a **tower dump**:

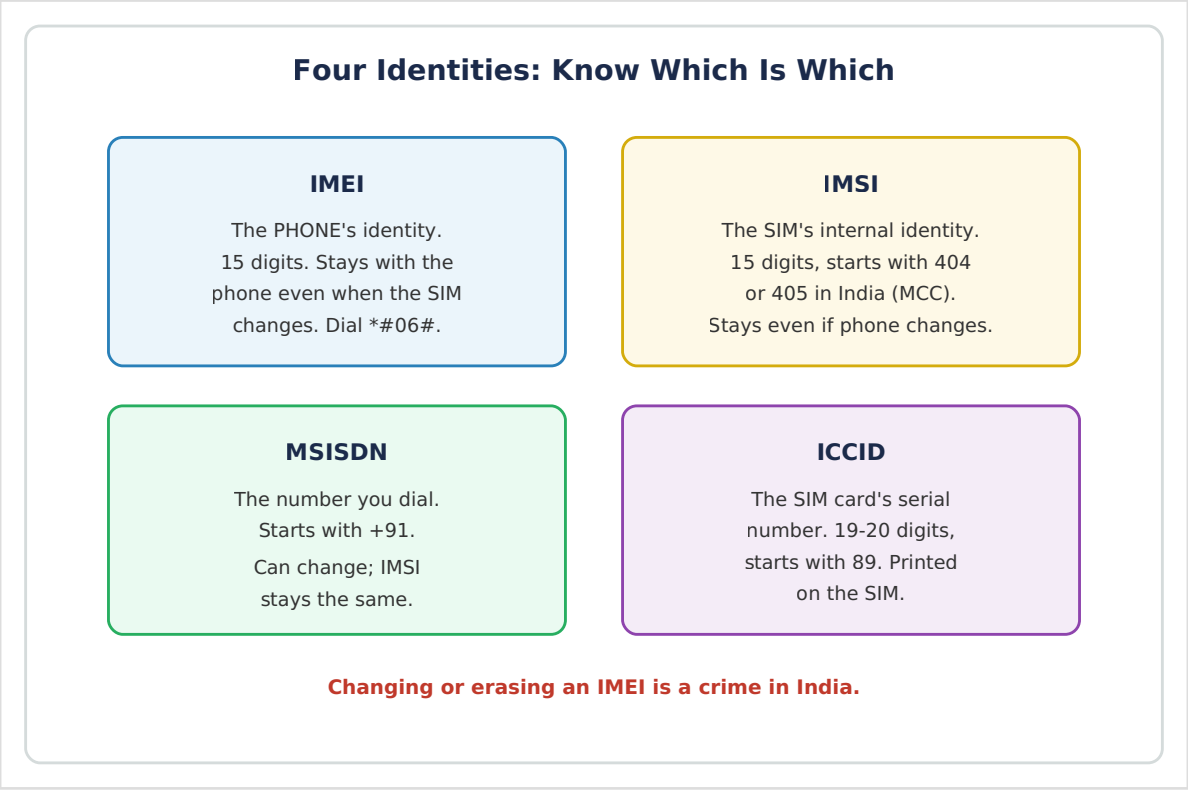
- All numbers on one tower in a **fixed time window** - the crime's place and hour
- A daily-seen number is likely a **resident**; a crime-window-only number is a **lead**
- The dump holds **innocents' numbers** - guard their privacy inside the case file
- Demanded only through **legal process** - and a window appearance is a lead, **never proof**

Filtering: From Noise to Leads



- Real work is **filtering**, not suspecting everyone
- Remove the daily-seen; **shortlist the window-only**
- Protect the innocent - **always**

Four Identities: Know Which Is Which



- IMEI is the **phone**; IMSI the **SIM**; MSISDN the **number**; ICCID the **SIM serial**
- Criminal changes SIM? **IMEI stays**. Changes phone? **IMSI stays**
- Dual-SIM phones carry **two IMEIs** - note both

IMEI: The Unchanging Identity

IMEI - International Mobile Equipment Identity - the phone's fingerprint. It **does not change** when the SIM changes:

- 15 digits - on the box, the bill, settings, or dial ***#06#**
- First step in any phone case: **handset identification by IMEI**
- Which SIMs lived in this IMEI? That question **reaches the criminal** through every SIM swap
- Stolen phone? **CEIR portal (ceir.gov.in)** blocks it on all networks - FIR plus Request ID, unblock when found

Changing an IMEI Is a Crime

Stolen phones get new IMEIs and new buyers. **The law treats that as a crime in India.**

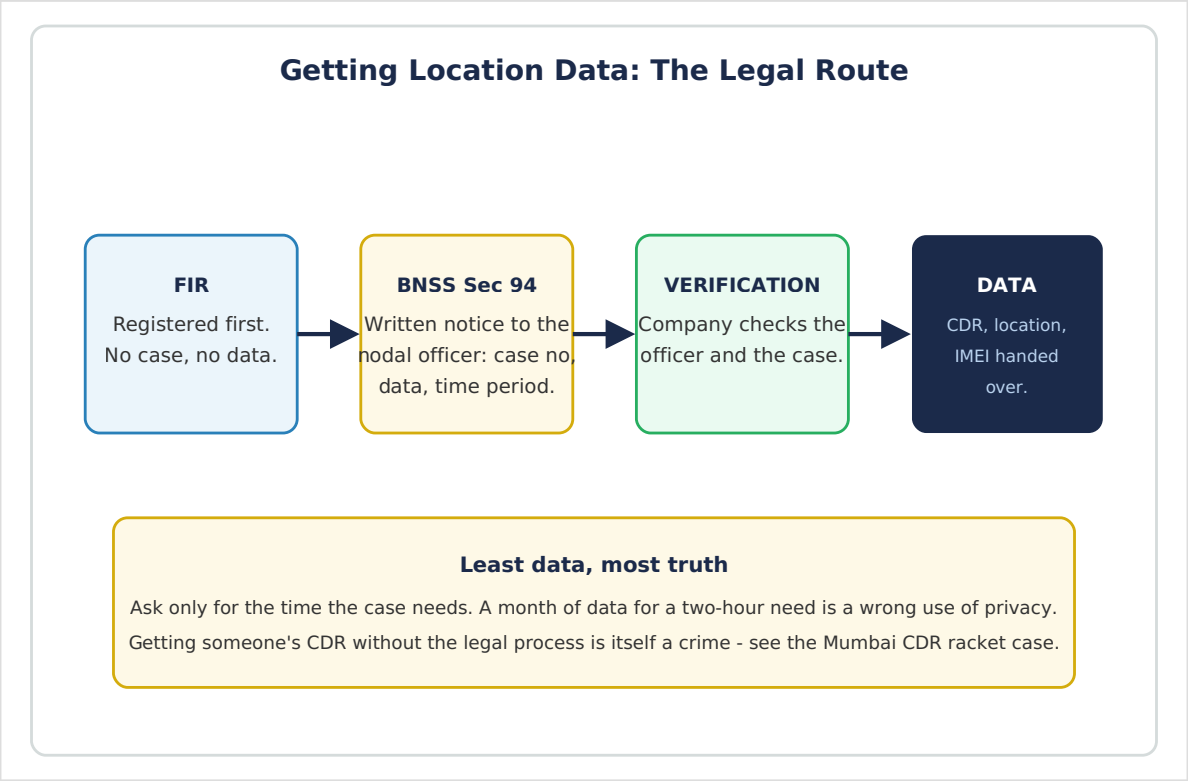
- IMEI-changing rackets **launder stolen phones** back into the market
- If a seized phone's IMEI looks **tampered**, note it - it widens the case
- Buyers of such phones become **part of the chain** - question them too
- Your own IMEI: **note it safely, share it with no stranger**

Worked Example: Number to Person

Join every link - **this is the method**, fictional numbers, real technique:

- Fraud call from an unknown number - first get its **CDR**: contacts, towers, timing
- The SIM's **IMSI** names the subscriber - often fake; the phone's **IMEI** gives the real clue
- Which SIMs **earlier lived** in this IMEI? Their owners' trail leads to the real user
- Number to tower, tower to location, SIM to IMSI, phone to IMEI, **IMEI to person**

Getting Location Data: The Legal Route



- Four steps: FIR, BNSS s.94, verification, data
- **No case, no data** - the process protects privacy
- Least data, **most truth**

The BNSS Section 94 Notice

Location data, CDR and IMEI details are **private information**. The notice is your lawful key:

- First: an **FIR must exist** - data follows the case, never precedes it
- Written request to the **nodal officer**: case number, exact data asked, **time period**
- The company **verifies** the officer and the case before releasing anything
- This friction is **by design** - in wrong hands, this data ruins lives

Location Data Inside the Phone

The phone itself is a **location diary** - if seized correctly:

- System logs, **map app history**, photo metadata - locations hide in many places
- **Airplane mode first** - no remote wipe; then a forensic copy, analysis on the copy
- The forensic expert extracts it - **you secure the phone** so extraction stays possible
- Seizure done wrong means the diary **burns before it is read**

Limits of Location Evidence

An honest investigator knows what location data **cannot** prove:

- A tower tells **where the phone was** - not whose hand held it
- GPS **errs** between tall buildings; signals get disturbed
- Always read it **with other evidence** - CCTV, witnesses, call records
- Location alone **convicts no one** - it is one link; strong cases need every link telling **one story**

Mistakes That Become Crimes

With location data, a shortcut is not just sloppy - **it is criminal**:

- **Buying CDRs** from touts - the Mumbai racket ended with constables arrested; the data is poisoned anyway
- Demanding a **month of data** for a two-hour need - a wrong use of privacy
- Declaring someone guilty on **CDR alone** - the lab's fictional 00777 was a lead, not a convict
- Sharing tower-dump data **outside the case file** - innocent people's privacy in your hands

Real Case: The Racket Selling CDRs Got Caught

Mumbai Crime Branch, Unit 12. A man on Instagram called himself **'Inspector Sanjay'** - promising to return lost phones, for money.

- He **illegally extracted** victims' CDR and location data - 9 accused, including **two serving constables** of Jharkhand and Bihar police
- Recovered: **50 CDRs, 520+ SDRs**, location data of 550+ numbers, 420+ IMEI details
- Source The Hindu
- Lessons: this data comes **only through legal process**; sellers and buyers are **both criminals**; a good investigator never shortcuts

Your Turn: Labs in This Module

Four labs - the first two on **fictional practice data**, clearly marked:

- **Lab 1:** Read a sample CDR - top 5 contacts, night pattern, flag the crime-window number
- **Lab 2:** Filter a tower dump - drop the daily-seen, shortlist, cross-check with the CDR
- **Lab 3:** Classify IMEI, IMSI, MSISDN, ICCID - then dial *#06# on your own phone
- **Lab 4:** Walk through the CEIR portal - block, unblock, status check (information only)

Key Takeaways

If you remember nothing else from Module 20:

- **Three routes** to location: GPS, tower, WiFi - read them together
- CDR in **three passes**: top contacts, night pattern, crime window
- **IMEI never changes** with the SIM - it is your most trustworthy identity
- Filter tower dumps; **protect the innocent**
- Data comes **only via BNSS s.94** - shortcuts are crimes
- Location is **one link** - never the whole chain

Review Questions

- 1. From the Lab 1 fictional CDR: which number is the top contact, which shows the night pattern, and which appears only in the crime window? What is your next step for each?
- 2. A tower dump shows 8 numbers; 5 are daily-seen. Explain your filtering and what you do with the remaining 3.
- 3. The criminal changed both SIM and phone. Which identity survives each change, and how does the investigation continue?
- 4. Draft the four points of a BNSS s.94 notice for a tower dump. Why must the time period be narrow?
- 5. Tower location puts the accused's phone at the scene. Give two reasons this alone cannot convict - and what would strengthen it.

NEXT: MODULE 21

Advanced Intelligence Techniques

Stay one step ahead of the criminal

Cyber Crime Investigator - Training Manual